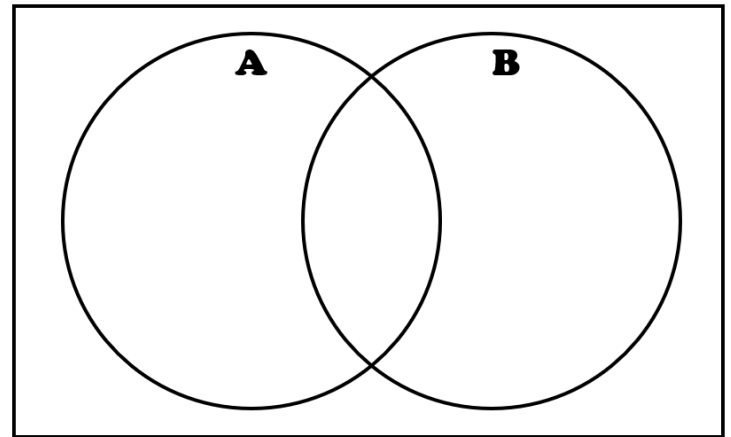


Venn Diagrams: Union and Intersection

- We can represent events with a Venn diagram
 - a display of potential probabilities.
- The box around the Venn diagram represents the total sample space where
- The circles themselves represent the probability of each event.



Union

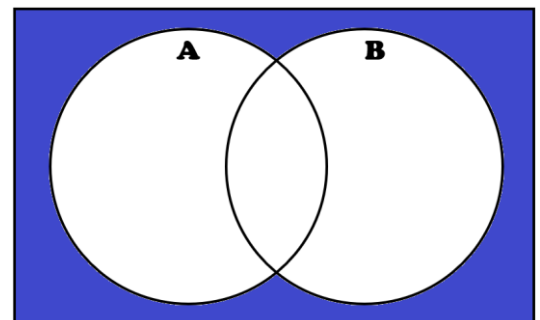
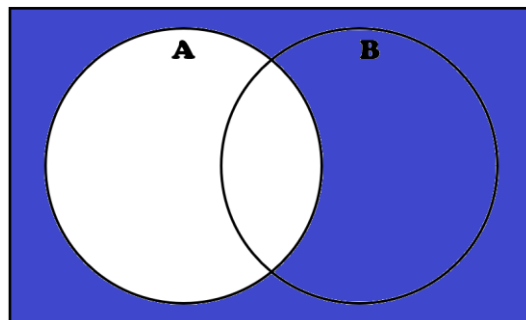
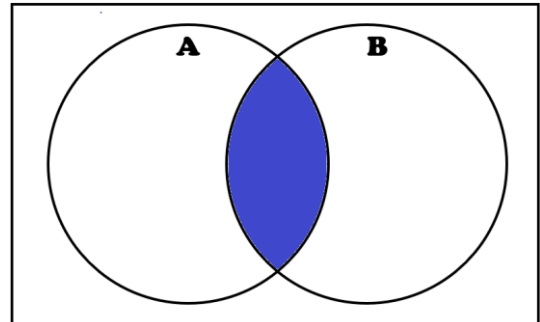
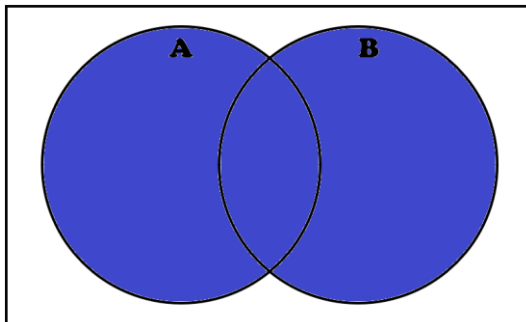
OR = _____

Symbol = _____

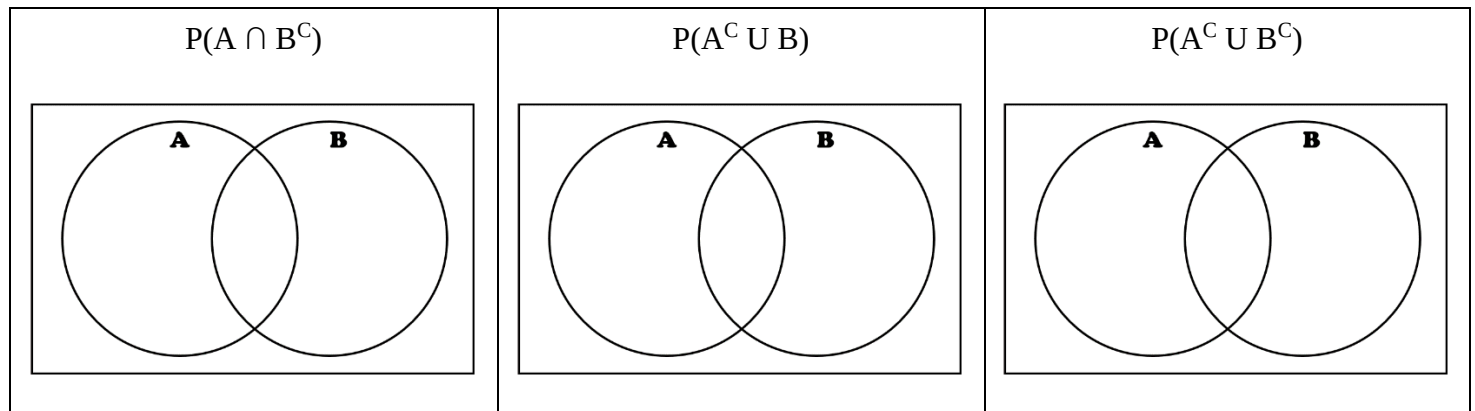
Intersection

AND = _____

Symbol = _____

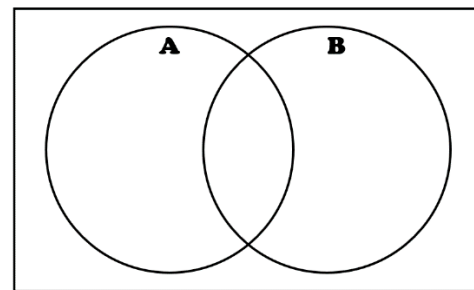


Example 1) Shade the probabilities for the following notations



Example 2) In your statistics class, you are interested in exploring the taste preference of your classmates. You have two tastes, sweet and salty. 25 students were asked if they prefer both, only sweet, only salty, or neither. The following two-way table shows the data.

	Yes Sweet	No Sweet	Total
Yes Salty	6	7	13
No Salty	11	1	12
Total	17	8	25



Let A = Sweet and B = Salty. Draw a Venn diagram in the space above and find and interpret the following probabilities.

a) $P(A \cap B)$

b) $P(A \cup B)$

c) $P(A \cap B^c)$

d) $P(A^c \cup B)$

e) $P(A \cup B^c)$

f) $P(A^c \cap B^c)$